

1. Administrative & Study Identification

Full Study Title: A Phase I, Open-label, Dose Finding Study of WVT078 Alone and in Combination with WHG626 in Patients Relapsed and/or Refractory Multiple Myeloma (MM) **Short Title/Acronym:** WVT078 Phase I in MM **Protocol Number:** CWVT078A12101 / NCI-2021-09189 **NCT Number:** NCT04123418 **Sponsor Name:** Novartis Pharmaceuticals Corporation **Investigational Product:** WVT078 (BCMA×CD3 bispecific antibody) alone and in combination with WHG626 (gamma-secretase inhibitor / AL102) **Phase of Development:** Phase I **Medical Monitor:** Novartis Medical Monitor

2. Study Synopsis & Rationale

2.1 Study Objective * **Primary Objective:** To characterize the safety, tolerability, and determine the Maximum Tolerated Dose (MTD) / Recommended Dose (RD) regimen(s) of WVT078 alone and in combination with WHG626 in subjects with relapsed and/or refractory multiple myeloma. * **Secondary Objectives:** To assess preliminary anti-MM response (efficacy), and characterize the pharmacokinetics (PK), pharmacodynamics (PD), and immunogenicity of WVT078 alone and in combination with WHG626.

2.2 Background & Rationale B-cell maturation antigen (BCMA) is an ideal target in multiple myeloma (MM) due to its highly specific expression in malignant plasma cells. WVT078 is an anti-BCMA×anti-CD3 bispecific antibody that redirects T-cells to target and eliminate MM cells. WHG626 is a gamma-secretase inhibitor (GSI) that prevents the cleavage of BCMA from the cell surface, potentially enhancing the efficacy of WVT078. This study is designed to evaluate these therapies in patients who have exhausted standard-of-care options.

3. Study Design & Methodology

3.1 Design Framework * **Study Type:** Interventional * **Control Type:** Single-arm (Dose-escalation) * **Blinding:** Open-label * **Randomization:** N/A

3.2 Planned Enrollment & Duration * **Target N:** 56 evaluable patients (based on dose-escalation completion data) * **Number of Sites:** Multicenter (Global) * **Estimated Study Start:** 2019 * **Estimated Primary Completion:** 2023/2024 (Primary dose-escalation phase)

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age 18 to 99 years. 2. Diagnosis of relapsed and/or refractory Multiple Myeloma (MM). 3. Relapsed and/or refractory to two or more regimens including an immunomodulatory drug (IMiD), a proteasome inhibitor, and an anti-CD38 agent (if available).

4.2 Exclusion Criteria 1. Use of systemic chronic steroid therapy (≥ 10 mg/day prednisone or equivalent) or any immunosuppressive therapy within 7 days of the first dose of study treatment. 2. Active known or suspected autoimmune disease or impaired cardiac function/clinically significant cardiac disease. 3. Active central nervous system involvement by malignancy or presence of symptomatic CNS metastases. 4. Treatment with cytotoxic or small molecule antineoplastics or any experimental therapy within 14 days or 5 half-lives, whichever is shorter.

5. Intervention & Treatment Plan

5.1 Investigational Regimen * **Dosage/Frequency:** WVT078 administered via dose-escalation (ranging from 3 to 250 μ g/kg) once weekly. In the combination arm, WHG626 is administered at 2-4 mg once daily, given 2 days on and 5 days off. * **Route of Administration:** Intravenous (IV) for WVT078; Oral for WHG626. * **Duration of Treatment:** Until disease progression or unacceptable toxicity.

5.2 Concomitant & Prohibited Therapy * **Permitted:** Standard supportive care medications for MM complications. * **Prohibited:** Immunosuppressive therapies, chronic systemic steroids (≥ 10 mg/day prednisone), and concurrent antineoplastic/experimental agents.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Baseline Tumour Assessment, Labs
Baseline	Day 0	First Dose, Safety Labs, PK/PD Baseline, ECG
Interim	Weekly / Per Cycle	IV Administration, DLT Assessment, Efficacy Assessment, CRS monitoring
End of Study	Treatment Discontinuation	Final Vital Signs, Exit Interview, IP Accountability, Follow-up

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint * **Definition:** Incidence of Dose-Limiting Toxicities (DLTs) to determine the Maximum Tolerated Dose (MTD)

and/or Recommended Dose (RD). * **Measurement Method:** Clinical and laboratory evaluation of adverse events based on standard toxicity criteria during the primary dose-escalation cycle.

7.2 Secondary & Exploratory Endpoints * Overall Response Rate (ORR) and Complete Response Rate (CRR). * Characterization of Pharmacokinetics (PK) and Immunogenicity. * Pharmacodynamic (PD) biomarkers (e.g., serum sBCMA levels, cytokine production).

8. Safety & Adverse Event Reporting

- **Adverse Event (AE) Monitoring:** Continuous monitoring by the investigator, with a specific focus on Cytokine Release Syndrome (CRS), neurotoxicity, and hematologic toxicities.
 - **Serious Adverse Events (SAEs):** Standard reporting timeline, typically within 24 hours of investigator awareness.
 - **Data Monitoring Committee (DMC):** Internal safety review committee utilized to make dose-escalation decisions.
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9. Statistical Analysis Plan (SAP) Summary

- **Analysis Populations:** Safety Set for all treated patients.
 - **Sample Size Justification:** Guided by the Bayesian optimal interval (BOIN) design for dose-escalation rather than strict power calculations.
 - **Handling of Missing Data:** Missing data will be handled via descriptive statistics without imputation for safety endpoints.
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10. Quality Assurance & Regulatory Compliance

- **GCP Compliance:** This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.
 - **Monitoring Plan:** Regular onsite and remote monitoring visits to ensure protocol adherence and patient safety.
 - **Audit Trail:** Details on eCRF locking, source data verification, and data clarification forms managed by the sponsor's data management team.
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11. Study Identifiers & Governance

- **Primary ID:** CWVT078A12101
- **Registry Number:** NCT04123418
- **Sponsor/Lead Organization:** Novartis Pharmaceuticals Corporation
- **Study Title:** A Phase I, Open-label, Multicenter, Study of WVT078 in Patients With Multiple Myeloma (MM)

- **Official Regulatory Title:** A Phase I, Open-label, Dose Finding Study of WVT078 Alone and in Combination with WHG626 in Patients Relapsed and/or Refractory Multiple Myeloma (MM)
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12. Clinical Framework (Multiple Myeloma Specific)

- **Primary Condition:** Relapsed and/or Refractory Multiple Myeloma (RRMM)
 - **Diagnosis Threshold:** Disease relapsed/refractory to ≥ 2 standard regimens (including an IMiD, proteasome inhibitor, and anti-CD38 agent)
 - **Medical Methodology:** T-cell redirecting therapy (Bispecific Antibody + Gamma-secretase inhibitor)
 - **Optimization Target:** BCMA and CD3 engagement resulting in malignant plasma cell depletion
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13. Study Procedures & Methodology

- **Management Pathway:** Stepwise dose-escalation guided by the BOIN design
 - **Education Protocol (HCP):** Site initiation and rigorous training on Cytokine Release Syndrome (CRS) grading and mitigation
 - **Education Protocol (Patient):** Informed consent processes and instructions on reporting immediate signs of systemic reactions
 - **Quality Audit Mechanism:** Continuous safety reviews between cohorts prior to dose-escalation approval
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14. Observation & Follow-Up Schedule

- **Total Duration:** Continuous until disease progression or unacceptable toxicity
 - **Onsite Visit Cadence:** Weekly visits required during initial cycles for IV administration and intense safety monitoring
 - **Remote Monitoring Frequency:** As per Novartis clinical monitoring plan
 - **Checklist Review Interval:** End of each treatment cycle for DLT and response assessment
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15. Sign-off & Approval

	Role		Name		Signature		Date			:---		:---		:---		:---	
	Principal Investigator			[Insert Name]		_____		[DD-MMM-YYYY]									
	Clinical Operations Lead			[Insert Name]		_____		[DD-MMM-YYYY]									
	Lead Statistician			[Insert Name]		_____		[DD-MMM-YYYY]									